

GAVLE, Sweden

WMO: 024530

Lat: 60.7161N

Lon: 17.1607E

Elev: 52

StdP: 14.67

Time Zone: 1.00 (EUC)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.7	3.6	-7.2	3.8	-1.7	-0.8	5.3	4.1	19.5	33.9	17.7	31.3	1.4	200	0.602

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.3	78.7	64.6	75.2	62.7	71.9	61.4	66.6	74.6	64.6	71.9	62.9	69.5	7.5	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.8	88.7	69.4	61.9	82.9	67.6	60.0	77.5	66.1	31.2	74.6	29.7	72.3	28.5	69.6	73.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
17.1	14.9	13.3	DB	-9.8	83.1	6.2	3.5	-14.3	85.7	-17.9	87.8	-21.4	89.8	-25.9	92.3	(4)
			WB	-9.7	69.3	5.7	2.6	-13.8	71.2	-17.1	72.7	-20.3	74.2	-24.5	76.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	42.1	25.2	26.4	31.3	39.5	48.1	55.9	60.7	59.4	52.2	42.3	34.8	28.6
	DBStd	14.45	10.11	9.67	8.47	6.37	6.61	5.49	4.75	4.96	5.61	6.55	7.13	9.59
	HDD50	3866	769	661	581	320	114	10	0	1	41	250	457	662
	HDD65	8387	1234	1081	1046	765	525	278	151	185	385	704	906	1127
	CDD50	987	0	0	0	5	55	186	331	292	106	12	0	0
	CDD65	34	0	0	0	0	1	4	18	10	1	0	0	0
	CDH74	411	0	0	0	0	19	61	207	121	3	0	0	0
	CDH80	49	0	0	0	0	0	6	30	13	0	0	0	0
Wind	WSAvg	6.0	5.9	6.2	6.4	6.3	6.6	6.2	5.6	5.3	5.5	5.8	6.1	6.1
Precipitation	PrecAvg	24.9	1.8	1.3	1.3	1.6	1.5	2.0	2.9	3.1	2.6	2.3	2.4	2.1
	PrecMax	34.8	4.7	3.9	2.9	3.9	4.6	7.4	6.5	10.0	8.4	6.7	5.7	5.6
	PrecMin	17.4	0.2	0.2	0.1	0.1	0.0	0.5	0.6	0.0	0.5	0.5	0.5	0.0
	PrecStd	4.7	1.0	0.7	0.7	1.0	1.1	1.3	1.6	2.1	1.6	1.5	1.2	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.4	47.0	55.8	67.1	76.9	80.3	83.9	81.6	73.3	61.2	53.1	49.5
		MCWB	40.8	41.5	45.1	51.9	59.4	63.6	67.1	65.1	61.6	54.7	49.8	46.5
	2%	DB	41.6	43.9	50.4	61.1	69.9	74.9	79.3	77.6	68.9	57.4	49.9	45.3
		MCWB	38.5	39.3	42.2	49.0	55.9	60.3	65.7	64.6	59.4	52.9	47.2	42.8
	5%	DB	39.7	41.1	46.6	55.9	65.4	71.2	75.8	73.6	65.3	54.8	47.1	42.7
		MCWB	37.3	37.7	39.9	46.6	53.7	59.1	63.8	63.0	57.8	51.4	44.9	40.4
	10%	DB	37.5	38.6	43.1	51.8	61.3	67.8	72.1	70.2	62.4	52.3	44.9	40.0
		MCWB	35.6	35.8	37.9	44.2	51.2	57.6	62.4	61.7	56.3	49.5	43.2	38.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	41.6	42.5	45.8	53.9	61.3	65.3	69.4	69.6	63.4	56.3	50.4	46.4
		MCDB	43.6	46.0	53.6	65.3	73.1	75.3	79.4	76.3	69.5	59.5	52.5	48.3
	2%	WB	39.3	40.1	43.1	50.0	57.3	62.6	67.2	66.5	60.8	54.0	47.4	43.0
		MCDB	41.0	42.9	49.2	59.4	66.7	70.8	76.2	73.2	66.4	56.4	49.5	44.9
	5%	WB	37.5	38.1	40.7	47.3	54.7	60.6	65.3	64.5	59.0	52.0	45.2	40.5
		MCDB	39.3	40.6	45.5	54.7	63.7	69.1	73.2	71.0	64.4	54.4	46.8	42.1
	10%	WB	35.8	36.0	38.3	44.7	52.2	58.7	63.4	62.8	57.2	49.8	43.4	38.2
		MCDB	37.5	38.2	42.5	50.7	60.1	66.1	70.5	68.6	61.3	52.0	44.7	39.7
Mean Daily Temperature Range	5% DB	MDBR	10.7	12.1	14.6	16.5	17.8	16.8	17.3	18.2	16.4	12.3	8.3	9.2
		MCDBR	9.2	12.3	18.2	25.7	27.0	24.1	24.6	24.2	20.7	14.8	11.1	9.4
		MCWBR	8.2	9.3	11.9	15.2	14.7	12.3	12.5	12.6	12.5	11.0	9.6	8.1
	5% WB	MCDBR	8.8	11.3	17.1	23.4	23.1	20.9	21.4	20.1	18.5	13.4	10.5	8.7
		MCWBR	8.1	9.0	11.7	14.4	13.6	11.6	12.3	11.9	12.1	10.6	9.3	8.0
Clear-Sky Solar Irradiance	taub	0.262	0.308	0.330	0.364	0.364	0.357	0.380	0.372	0.339	0.325	0.285	0.366	
	taud	2.189	2.280	2.358	2.353	2.417	2.472	2.428	2.444	2.523	2.439	2.240	2.655	
	Ebn at Noon	156	215	252	264	273	277	266	259	250	210	143	109	
	Edh at Noon	12	21	27	32	32	31	32	29	23	18	12	8	
All-Sky Solar Radiation	RadAvg	84	286	757	1223	1564	1741	1621	1260	816	370	111	51	
	RadStd	11	51	89	117	163	124	164	127	103	61	26	6	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (44 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	-180	N/A	N/A	N/A

Nomenclature: See separate page